

# Lineage and Release Timing in Correlated Errors of Open-Weight Language Models: A Precision-Gated, Pre-Registered Study of 977 Models

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**ABSTRACT** Ensembles and review pipelines built from several open-weight language models assume that their members err somewhat independently. When two models share an ancestor, or were released at the same time, that assumption may fail. We ask which of the two is associated with models choosing the *same wrong answer*. We first show why the question cannot be answered by decomposing per-model accuracy into family and release-era variance components: a precision analysis of the fitted random-effects model shows that, with five to seven model families and at most fourteen release quarters, neither share can be estimated to better than about  $\pm 18$  percentage points (root-mean-square error), whatever the number of models. We therefore move to a pair-level design. Using item-level predictions for 977 models from the Open LLM Leaderboard (MMLU, 14,042 items), validated item by item against the leaderboard's stored correctness, we resolve each model's lineage root from Hugging Face `base_model` declarations and relate pairwise agreement on wrong answers to shared lineage and to the gap between upload months. The analysis, sample rules and a simulation-based power gate were fixed before outcomes were computed; three dated amendments are disclosed. In the pre-registered primary analysis (590 models, 173,755 pairs), sharing a lineage root is associated with a 13.3 percentage-point higher probability of choosing the same wrong option when both models err (95% CI 11.8–14.8; delete-one-root jackknife CI 10.8–15.8), against a mean of 42.3%. The estimate is 11.7–13.4 points across all twelve pre-registered analyses and stays between 10.2 and 14.2 points under root-level inference, alternative weighting, a position-adjusted outcome and exploratory size controls. Item-difficulty-aware null models absorb almost all of the average agreement but leave the lineage contrast unchanged. Release-month proximity shows no association of consistent sign with the choice of wrong option: the same-month coefficient is +0.004 (CI  $-0.003$  to  $+0.011$ ) in the primary analysis, small and positive under some weightings, and negative ( $-0.015$ , CI  $-0.027$  to  $-0.002$ ) among models scoring at least 30%; for which items fail together, a small positive association appears in the full samples but not among models above 30%. The associations are observational and describe a population dominated by community fine-tunes; they do not separate the pretraining era of foundation models from their lineage.

**INDEX TERMS** Large language models, correlated errors, model lineage, ensembles, dyadic regression, pre-registration, Open LLM Leaderboard

## I. INTRODUCTION

WHEN several public language models are asked the same hard question, they often fail together, and often in the same way. Kim *et al.* [1] document this across hundreds of open and hosted models: when two models are both wrong,

they choose the same wrong option far more often than chance. For anyone who combines models—a majority vote, a routing scheme, or a review step that escalates only when models disagree—this matters directly. Agreement between members is informative only if their errors are not shared [18],

[20].

Two explanations for shared errors compete. *Lineage*: a model fine-tuned from another inherits its parameters and therefore, plausibly, its blind spots. *Era*: models built at the same time draw on the same data, recipes and fashions and may acquire the same mistakes without any shared ancestry. The two are entangled, because descendants are always released after their ancestors. Which one dominates has a practical consequence: if lineage dominates, combining models from different base models should reduce correlated error; if era dominates, it may help little.

This paper reports a study that answers a narrower, well-posed version of that question, and the methodological path that led to it. An earlier version of this work attempted to decompose per-model benchmark accuracy into family and release-era variance components, applied a design “gate” to sixteen evaluated models, and found that the gate failed. On re-examination that version had three defects: its gate tested the rank of a fixed-effects design that the fitted random-effects model does not use; the evaluation code that produced its per-question predictions recorded only the first answer option, making all item-level outputs constant; and the variance decomposition, even when identified, cannot be estimated precisely with the handful of families and quarters available. Section III documents the last point, which motivates the pair-level design used here. Appendix A records what changed from the earlier version.

## A. RESEARCH QUESTIONS

- 1) **RQ1:** Can family and release-era variance shares of per-model accuracy be estimated precisely from realistic model populations?
- 2) **RQ2:** Among open-weight models, is sharing a lineage root associated with choosing the same wrong answer, at equal release-time gap?
- 3) **RQ3:** Is release-time proximity associated with choosing the same wrong answer, at equal lineage status?

## B. CONTRIBUTIONS

- 1) **A precision analysis for the per-model design.** We identify the correct identifiability condition for the crossed random-effects model and show by expected-information and Monte Carlo analysis that family and era shares are imprecise with the number of families and quarters that real populations offer (Section III).
- 2) **A pre-registered pair-level design with a simulation gate.** Agreement on wrong answers is modelled at the level of model pairs, with lineage and release-month gap as pair covariates and dyadic cluster-robust inference. Sample rules were fixed by a simulation-based power and separation gate before outcomes were read (Section IV).
- 3) **A validated item-level dataset.** We extract item-level predictions for 977 leaderboard models, validate every item against the leaderboard’s stored correctness and reproduce official aggregate scores, and resolve lin-

age from model-card declarations with a separately reported, lower-confidence extension (Section V).

- 4) **Results with explicit robustness.** Shared lineage is strongly and consistently associated with shared wrong answers; release-month proximity is not, in a consistent direction (Section VI).

## II. RELATED WORK

**Correlated errors in LLMs.** Kim *et al.* [1] measure pairwise agreement on wrong answers across 349 Hugging Face models and 71 HELM models and regress it on pair characteristics, finding that a shared provider, a shared base architecture and similar size predict higher agreement, and that larger and more accurate models have more correlated errors even across architectures and providers. They suggest that newer models may be converging in their outputs but do not test this over release time. Chen [2] shows, across 67 models, that the gain from routing, voting or mixture-of-agents is capped by the rate at which all models fail the same query, and that average pairwise error correlation understates that rate. Kuai *et al.* [8] audit 18 models from six families for synchronized failures with difficulty-weighted dependence metrics, relate dependence to judges over-endorsing responses, and reweight verifier ensembles accordingly. Goel *et al.* [3] propose a chance-adjusted similarity metric based on overlapping mistakes and report that mistakes become more similar as capability increases. Our temporal predictor is different from both accuracy-related correlation and model age: it is the *proximity* of two models’ release months, estimated at equal lineage status and conditional on both models’ accuracies, which the model controls for (the accuracy-sum coefficient is positive, consistent with [1]). We make no claim about absolute model age.

**Null models for excess agreement.** Jo, Garg and Raghavan [23] argue that whether models agree “more than expected” depends on two analyst choices: the null model of independence, where accounting for item difficulty leads to very different conclusions from nulls that do not, and the population of models and items compared. They conclude that monoculture is a context-dependent inference rather than an absolute property of models. This bears directly on our study, which uses one benchmark and one leaderboard population. Three estimands are in use: correlated correctness (whether two models fail the same items, e.g. the phi coefficient of error indicators, or co-failure in excess of an item-difficulty null); conditional agreement on the wrong option (whether two models that both fail an item choose the same distractor [1]); and accuracy-adjusted similarity [3]. Our primary estimand is the second and our secondary the first. More importantly, our quantity of interest is a *contrast*—the difference in agreement between pairs that share a lineage root and pairs that do not, at equal release-month gap—not the *level* of excess agreement that the null-model debate concerns. A null that shifts all pairs’ expected agreement equally leaves such a contrast unchanged; a null that differs between related and unrelated pairs does not. Section VI-D therefore re-estimates

both outcomes against item-difficulty-aware nulls.

**Our contribution.** We do not claim to be the first study of correlated errors in language models. To our knowledge, and based on the work reviewed here, this is the first study to examine declared fine-tune ancestry and release-month proximity jointly as correlates of conditional same-wrong-answer agreement. Relative to the work above, the outcome is conditional same-wrong-option agreement between pairs of models (not the all-fail rate of an ensemble [2], judge dependence [8] or pipeline measurement error [13]); lineage is each model's declared fine-tune ancestry resolved to a root checkpoint (not provider or architecture labels [1], family membership [8] or dataset lineage [7]); and release time enters as the proximity of two models' upload months, estimated jointly with lineage. We separate the two explicitly: lineage is taken from declared ancestry rather than provider labels, release time is the gap between upload months in categorical bins, inference accounts for the dependence of pairs that share a model or a lineage root, and the design was checked by a pre-specified simulation gate.

**Lineage and ancestry.** Laufer, Oderinwale and Kleinberg [4] map the family trees of about 1.86 million Hugging Face models from their declared parents and study how metadata traits—licenses, language support, model cards—are inherited and drift along lineages. Horwitz, Shul and Hoshen [5] recover model trees from model weights alone. Both treat lineage as the object of study; neither measures whether related models make the same errors on benchmark items. PhyloLM [6] reconstructs phylogenies of language models from output similarity; Li *et al.* [7] reconstruct the lineage of post-training *datasets* and show that benchmark contamination can propagate along it. These works infer or exploit lineage of models or data; we take lineage from declarations and ask how much error agreement it accounts for relative to release time.

**Variance components and dyadic inference.** Crossed variance-component models estimated by restricted maximum likelihood [9]–[11] underlie the per-model analysis in Section III; generalizability theory [12] asks the same question of measurement facets, and Messing [13] shows that ignoring design choices in LLM pipelines (judge model, temperature, prompt) makes evaluation uncertainty, including on MMLU, look smaller than it is. Pairwise outcomes are not independent, because every model appears in many pairs; we use dyadic cluster-robust variance estimators [14]–[16] and a delete-one-root jackknife [17].

**Ensembles and monoculture.** Ensemble gains depend on error diversity [18]–[20]; algorithmic monoculture arguments [21], [22] treat correlated error as an input. Our results bear on which kind of diversity is associated with less shared error, without testing ensembles directly.

### III. WHY PER-MODEL VARIANCE DECOMPOSITION IS NOT ENOUGH

#### A. MODEL AND IDENTIFICATION

Let  $Y_i$  be a per-model trait (e.g., accuracy), with family  $f(i)$  and release quarter  $e(i)$ :

$$Y_i = \mu + \alpha_{f(i)} + \beta_{e(i)} + u_i, \\ \alpha \sim \mathcal{N}(0, \sigma_L^2), \quad \beta \sim \mathcal{N}(0, \sigma_E^2), \quad u \sim \mathcal{N}(0, \sigma_U^2), \quad (1)$$

so that  $\text{Var}(\mathbf{y}) = \sigma_L^2 \mathbf{Z}_F \mathbf{Z}_F^\top + \sigma_E^2 \mathbf{Z}_E \mathbf{Z}_E^\top + \sigma_U^2 \mathbf{I}$ , with only the intercept as a fixed effect. The variance components are identified when the covariance basis  $\{\mathbf{Z}_F \mathbf{Z}_F^\top, \mathbf{Z}_E \mathbf{Z}_E^\top, \mathbf{I}\}$  is linearly independent. The rank of the fixed-effects dummy design  $[\mathbf{1} \mid \mathbf{Z}_F \mid \mathbf{Z}_E]$ , used as the gate in the earlier version of this work, is not the relevant condition for this model: for the sixteen models evaluated there, that design is rank-deficient (rank 14 of 15) while the covariance basis has full rank (3 of 3). Its condition number also depends on the arbitrary choice of reference level (TABLE I).

#### B. PRECISION

Identification does not imply precision. For each design we compute the expected restricted-likelihood information  $I_{jk} = \frac{1}{2} \text{tr}(\mathbf{P} \mathbf{V}_j \mathbf{P} \mathbf{V}_k)$  at four share scenarios (lineage-dominant 0.5/0.2/0.3, era-dominant 0.2/0.5/0.3, balanced, and low-signal 0.1/0.1/0.8), the delta-method standard error of each share, and, as the verdict, the Monte Carlo root-mean-square error (RMSE) of the share estimates under the same REML estimator (500 replicates per scenario). The analytic standard error tracks the Monte Carlo RMSE closely (e.g., 0.289 versus 0.285 for the sixteen-model design), so it can be used before any measurement.

No design reaches an RMSE of 0.10 for both shares (TABLE I): not the sixteen evaluated models (0.30), not the 47-model candidate frame (0.19), and not the 30-model configuration the earlier version described as sufficient (0.19). Removing the single model responsible for the fixed-effects rank deficiency leaves the RMSE unchanged (0.302). The constraint is the number of *levels* of each factor, not the number of models. In staggered designs (Fig. 1), bringing both shares within  $\pm 0.10$  requires on the order of sixty families and fourteen quarters; with eight quarters the era share never falls below about 0.14. Because release quarters are few by construction, a per-model decomposition is a poor instrument for RQ2–RQ3. This answers RQ1 in the negative and motivates the pair-level design.

### IV. PAIR-LEVEL DESIGN

#### A. OUTCOME

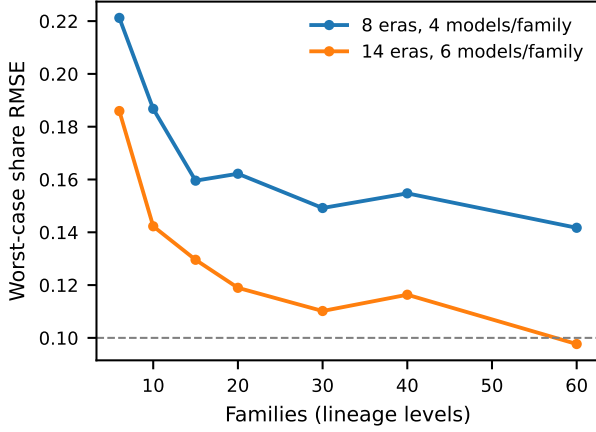
For models  $i$  and  $j$  evaluated on the same  $K = 14,042$  MMLU items [24], let  $c_{ik}$  be model  $i$ 's chosen option on item  $k$  (the argmax of its per-option log-likelihoods) and  $g_k$  the correct option. The outcome is the conditional agreement [1]

$$a_{ij} = \frac{\sum_k \mathbf{1}[c_{ik} = c_{jk} \neq g_k]}{\sum_k \mathbf{1}[c_{ik} \neq g_k] \mathbf{1}[c_{jk} \neq g_k]}, \quad (2)$$

the probability of choosing the same wrong option given that both are wrong. Every pair contributes all items on which both

**TABLE I: Precision of family and era variance shares for per-model traits.** FE: fixed-effects dummy design used as a gate in the earlier version;  $\kappa$  range is over every choice of reference levels. Covariance-basis rank 3/3 means the variance components are identified. The last column is the worst-case Monte Carlo RMSE of either share across four scenarios (500 replicates each).

| Design                       | Families | Quarters | FE full rank | $\kappa(X^\top X)$ range | Cov. basis | Worst RMSE     |
|------------------------------|----------|----------|--------------|--------------------------|------------|----------------|
| 16 evaluated models          | 5        | 11       | no           | singular                 | 3/3        | 0.302          |
| 22-model selection           | 6        | 14       | yes          | 285–1526                 | 3/3        | 0.256          |
| 47-model candidate frame     | 6        | 14       | yes          | 166–1116                 | 3/3        | 0.189          |
| 30-model “sufficient” design | 6        | 8        | yes          | 91–144                   | 3/3        | 0.194          |
| 42-model sweep design        | 7        | 8        | yes          | 83–113                   | 3/3        | 0.182          |
| nested reference (72)        | 6        | 6        | no           | singular                 | 2/3        | not identified |



**FIGURE 1: Precision of per-model variance shares versus the number of families.** Worst-case Monte Carlo RMSE of the family or era share across four scenarios (300 replicates each) for staggered designs. The dashed line marks  $\pm 0.10$ .

models are wrong; in the data this denominator ranges from 1,166 to 10,820 items (median 4,109 in the primary sample), so every  $a_{ij}$  is estimated from more than a thousand items.

### B. MODEL AND INFERENCE

$$a_{ij} = b_0 + b_L s_{ij} + \sum_g b_{E,g} \mathbf{1}[|m_i - m_j| \in g] + c_1(\text{acc}_i + \text{acc}_j) + c_2|\text{acc}_i - \text{acc}_j| + \varepsilon_{ij}, \quad (3)$$

where  $s_{ij}$  indicates a shared lineage root,  $m_i$  is the model’s upload month, and the gap bins are 0, 1–2, 3–5 and 6–11 months with 12 or more as reference.  $b_L$  is the association of shared lineage with agreement at equal time gap;  $b_{E,0}$  is the association of same-month release at equal lineage status. Accuracy terms control for the fact that weaker models have more, and different, errors. Because every model enters many pairs, pair residuals are dependent; the pre-registered standard errors are dyadic cluster-robust [14]–[16], allowing arbitrary correlation between any two pairs that share a model. Section VI-B adds root-level dyadic clustering and a delete-one-root jackknife.

### C. SIMULATION GATE AND AMENDMENTS

Before outcomes were read, the design was checked by simulation on the actual roster (roots and months only). Item-level choices were generated with wrong answers drawn from a

**TABLE II: Pair-level simulation gate (primary roster, 14,042 simulated items, 20 replicates per truth).** Null: largest rejection rate under no effect. Leakage: largest cross-term rejection. Power: rejection of the lineage / era term at the stated effect weight.

| Cap  | $N$ | Null | Leakage | Power $\lambda=0.05$ | Power $\lambda=0.03$ |
|------|-----|------|---------|----------------------|----------------------|
| 15   | 561 | 15%  | 20%     | 100% / 100%          | 50% / 75%            |
| 40   | 613 | 5%   | 10%     | 100% / 100%          | 40% / 95%            |
| none | 721 | 10%  | 15%     | 100% / 100%          | 95% / 95%            |

root-specific attractor (weight  $\lambda_L$ ), a month-specific attractor drifting over time ( $\lambda_E$ ), a global attractor, or uniform noise, and Eq. (3) was refitted. The pre-registered pass criteria were: null rejection at most 10%; cross-leakage (the lineage term rejecting under era-only truth, or vice versa) at most 10%; and power at least 80% for  $\lambda = 0.05$  on both terms. TABLE II shows the gate on the primary roster for three caps on the number of models per lineage root.

Three amendments were made and dated in the project’s pre-registration document, each before the quantities it concerns were computed: (1) the primary data source moved from the second to the first version of the leaderboard, whose item-level files are not access-gated; (2) the per-root cap was set to 40, the only setting meeting all gate criteria (TABLE II), and the gate was re-run with the real item count after an initial run with 1,500 items understated power; (3) after the per-model accuracies of the first 443 validated files showed many near-chance models, two sensitivity analyses were added before any pair outcome was computed: S1 adds the total-variation distance between two models’ distributions over chosen answer options as a covariate (an answer-option-distribution control: two models that both favor the same letters agree more for that reason alone, whatever their lineage), and S2 restricts to models with accuracy  $\geq 0.30$ .

## V. DATA

### A. SOURCE AND VALIDATION

The first version of the Open LLM Leaderboard [25] evaluated thousands of open-weight models on MMLU (5-shot) with the LM Evaluation Harness [26] and published per-item details, including per-option log-likelihoods. For each model we select the most complete run covering all 57 subjects and read four fields per item: the per-option log-likelihoods, the gold option, the leaderboard’s stored per-item correctness and, where present, an item hash. We accept the model only

**TABLE III: Model flow.** Exclusion counts refer to the frozen samples.

| Stage                                                 | Primary | Expanded |
|-------------------------------------------------------|---------|----------|
| v1 leaderboard models (contents table, de-duplicated) | 6,896   | 6,896    |
| eligible (primary lineage rules, dated)               | 721     | –        |
| eligible (expanded lineage rules, dated)              | –       | 1,295    |
| frozen download sample (cap 40 per root)              | 613     | 967      |
| validated answers (analysed)                          | 590     | 928      |
| of which root verified pretrained (strict)            | 554     | 789      |
| excluded: no complete MMLU run                        | 17      | 32       |
| excluded: details repository missing                  | 5       | 6        |
| excluded: unreadable predictions                      | 1       | 1        |

if (i) all 14,042 items are present, (ii) the argmax of the per-option scores reproduces the stored per-item correctness on *every* item, and (iii) item order and gold answers match a reference model. Three storage layouts occur; in the newest, the gold column is empty and gold is taken from the reference by position, which check (ii) then validates (a misaligned gold vector fails on roughly three quarters of items). Extracted macro-accuracy equals the leaderboard’s official results files for the models we spot-checked (43.80, 70.06 and 75.56). For each accepted model we store the chosen option (the argmax), the gold option, the item identifier and hash, the run identifier and the file layout; the per-option scores are used for validation and not retained.

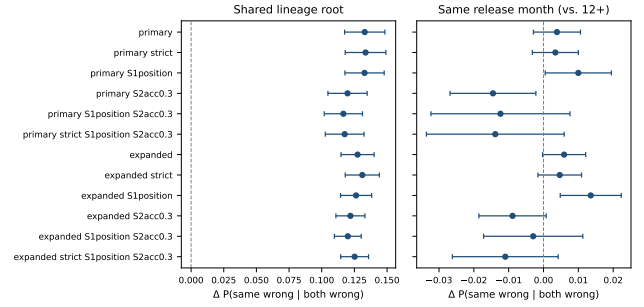
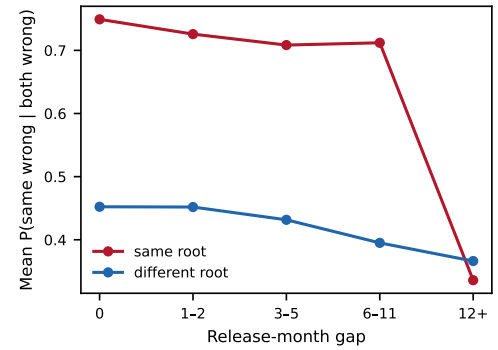
### B. LINEAGE AND RELEASE TIME

Lineage is resolved by following each model’s declared `base_model` on Hugging Face to a root, collapsing repository renames and a short, documented list of straight re-uploads and quantisations. A chain is unresolved if it passes through a merge (several parents), reaches a deleted or invalid ancestor, or ends at a fine-tune with no declared parent. Two populations are frozen: *primary*, in which every link comes from a model card, and *expanded*, which additionally accepts a model’s own first link from the `_name_or_path` field of its configuration file when that field names an existing Hugging Face repository (valid for 1,066 of 2,978 models lacking a card declaration, of which 574 then resolved to an eligible root; local paths, merge-tool paths and self-references were rejected). A *strict* variant keeps only chains ending at a checkpoint the leaderboard lists as pretrained. Release time is the Hub creation month of the model repository (27 distinct months, March 2022 to July 2024).

### C. MODEL FLOW

TABLE III reconciles the populations. The two frozen samples (at most 40 models per root, fixed seed) overlap in 562 models and together comprise 1,018; 977 passed validation and 41 were excluded and documented (33 without a complete MMLU run, 7 whose details repository is missing, and 1 with unreadable predictions). No excluded model was replaced.

The analysed primary sample comprises 590 models from 351 lineage roots, of which 63 roots contribute two or more

**FIGURE 2: Shared-root and same-month coefficients across the twelve pre-registered analyses (95% CIs, dyadic cluster-robust).****FIGURE 3: Unadjusted mean agreement on wrong answers by release-month gap, for pairs sharing a lineage root and other pairs (primary sample).**

models; the expanded sample comprises 928 models from 422 roots (108 with two or more). The largest roots are Llama-3-8B and Mistral-7B-v0.1 (39 and 37 models in the primary sample). Median accuracy is 0.415; 247 of 590 primary models score below 0.30, close to chance for four options, which motivated S1 and S2.

## VI. RESULTS

### A. PRE-REGISTERED ANALYSES

The mean agreement is 0.423 (primary). Kim *et al.* report the same mean for their Hugging Face population [1], but that population comes from a later leaderboard snapshot with a different question set (12,032 questions), so the coincidence is not evidence of validity. Unadjusted, pairs sharing a root agree on 73.1% of jointly wrong items and other pairs on 42.0% (Fig. 3). TABLE IV and Fig. 2 report all twelve pre-registered analyses.

**Lineage (RQ2).** In the primary analysis, sharing a root is associated with a 13.3-point higher agreement (95% CI 11.8–14.8). Across the twelve analyses the estimate ranges from 11.7 to 13.4 points; every confidence interval excludes zero by a wide margin. Controlling for the answer-option distribution (S1) leaves it unchanged; restricting to models above 0.30 accuracy (S2) reduces it slightly, to about 12 points.

**TABLE IV: Pre-registered analyses.** Coefficients of Eq. (3) with 95% confidence intervals from dyadic (model-level) cluster-robust standard errors. “Same month” is the 0-month gap bin relative to 12 or more months;  $p$  refers to it.

| Analysis                            | Models | Pairs   | Shared root             | Same month                 | $p$ (same month) |
|-------------------------------------|--------|---------|-------------------------|----------------------------|------------------|
| Primary                             | 590    | 173,755 | +0.133 [+0.118, +0.148] | +0.0039 [-0.0029, +0.0106] | 0.263            |
| Primary, verified roots             | 554    | 153,181 | +0.134 [+0.118, +0.149] | +0.0034 [-0.0032, +0.0100] | 0.313            |
| Primary + S1 (option distribution)  | 590    | 173,755 | +0.133 [+0.118, +0.148] | +0.0100 [+0.0005, +0.0195] | 0.039            |
| Primary, S2 (acc $\geq$ 0.30)       | 343    | 58,653  | +0.120 [+0.105, +0.135] | -0.0145 [-0.0268, -0.0022] | 0.021            |
| Primary + S1, S2                    | 343    | 58,653  | +0.117 [+0.102, +0.131] | -0.0124 [-0.0323, +0.0076] | 0.225            |
| Primary, verified + S1, S2          | 319    | 50,721  | +0.118 [+0.103, +0.132] | -0.0138 [-0.0336, +0.0059] | 0.170            |
| Expanded                            | 928    | 430,128 | +0.127 [+0.115, +0.140] | +0.0059 [-0.0003, +0.0121] | 0.061            |
| Expanded, verified roots            | 789    | 310,866 | +0.131 [+0.118, +0.144] | +0.0047 [-0.0016, +0.0109] | 0.145            |
| Expanded + S1 (option distribution) | 928    | 430,128 | +0.126 [+0.114, +0.138] | +0.0136 [+0.0048, +0.0223] | 0.002            |
| Expanded, S2 (acc $\geq$ 0.30)      | 589    | 173,166 | +0.122 [+0.111, +0.133] | -0.0089 [-0.0185, +0.0008] | 0.071            |
| Expanded + S1, S2                   | 589    | 173,166 | +0.120 [+0.110, +0.130] | -0.0030 [-0.0172, +0.0113] | 0.683            |
| Expanded, verified + S1, S2         | 481    | 115,440 | +0.125 [+0.115, +0.136] | -0.0110 [-0.0261, +0.0042] | 0.157            |

**Release time (RQ3).** In the primary analysis the same-month coefficient is +0.004 (CI -0.003 to +0.011,  $p = 0.26$ ). It is positive and nominally significant when the answer-option distribution is controlled (+0.010, CI +0.0005 to +0.0195,  $p = 0.04$ ), and negative when near-chance models are excluded (-0.015, CI -0.027 to -0.002,  $p = 0.02$  in the primary S2 analysis; -0.009,  $p = 0.07$  in the expanded S2 analysis). The 1–2, 3–5 and 6–11-month bins show the same pattern. The data therefore do not show a release-time association of consistent sign; where coefficients are nominally significant they are an order of magnitude smaller than the lineage association and change sign between specifications. This is a lack of consistent evidence, not evidence that the association is zero.

## B. DEPENDENCE AND INFERENCE

The 173,755 primary pairs are generated by 590 models, and pairs within a lineage root may be dependent beyond shared membership. TABLE V compares the pre-registered model-level dyadic standard errors with root-level dyadic clustering, which allows correlation between any two pairs touching the same root, and a delete-one-root jackknife, which removes all pairs involving one root at a time. Root-level procedures roughly double the standard error of the lineage coefficient; its intervals remain far from zero (primary: jackknife CI 10.8–15.8). For the same-month coefficient in the primary S2 analysis, the jackknife interval includes zero ( $p = 0.07$ ), so that negative estimate is not robust to root-level dependence.

## C. OUTCOME DEFINITION AND WEIGHTING

TABLE VI varies the outcome and weighting. Weighting pairs by their number of jointly wrong items raises the lineage estimate slightly (14.2 points) and makes the same-month coefficient positive (+0.012,  $p = 0.02$ ). The pre-registered secondary outcome, the phi correlation of error indicators, measures whether two models fail the *same items* rather than choose the same wrong option; it shows a lineage association of +0.170 and a smaller, positive same-month association (+0.018, CI +0.010 to +0.027). Subtracting the agreement expected from each model’s marginal distribution over answer options (exploratory; an item-level version follows in

Section VI-D) leaves the lineage estimate at 12.5 points and the same-month coefficient at -0.002. Restricting to pairs with at least 1,000 jointly wrong items changes nothing, because every pair already exceeds that threshold. A small, positive association between release proximity and *which items* fail together is therefore plausible; an association with *which wrong option* is chosen is not supported once the answer-option distribution is accounted for.

## D. ITEM-DIFFICULTY-AWARE NULLS

Added after the results were known, in response to review and following [23], these analyses are exploratory. Two nulls condition on the item.  $N1$  sets the expected agreement on each item to  $q_k = \sum_o p_k(o)^2$ , where  $p_k(o)$  is the share of wrong models choosing wrong option  $o$  on item  $k$ , so that an item with one attractive distractor is expected to produce agreement regardless of lineage; the outcome is observed agreement minus the mean of  $q_k$  over the pair’s jointly wrong items.  $N1a$  computes  $p_k$  from all models, which is conservative because a large lineage cluster pulls  $p_k$  towards its own choices;  $N1b$  weights every lineage root equally.  $N2$  fits a Rasch model,  $\text{logit } P(\text{wrong}_{ik}) = d_k - \theta_i$ , by joint maximum likelihood and measures co-failure in excess of  $\sum_k \hat{P}_{ik} \hat{P}_{jk}$ , per item. TABLE VII reports both with delete-one-root jackknife intervals.

Item-level distractor popularity accounts for almost all of the *average* agreement: the mean excess falls from 0.423 to 0.002 ( $N1a$ ) and 0.019 ( $N1b$ ), consistent with the observation of [23] that levels of excess agreement depend on the null. The lineage *contrast* does not change: pairs sharing a root exceed the item-aware expectation by 13.3 points ( $N1a$  and  $N1b$ , primary), and co-fail on 4.0% more items than the Rasch model predicts. The same-month coefficient for the choice of wrong option is unchanged (+0.004, jackknife  $p = 0.37$ ). For co-failure, a small positive same-month association remains in the full samples (+0.0036 per item, jackknife  $p = 0.003$ ) but not among models scoring at least 30% ( $p = 0.56$  and 0.48).

**TABLE V: Inference audit.** <sup>†</sup>Pre-registered. Root-level procedures treat lineage roots as the unit of dependence.

| Sample       | Term       | Standard error            | Estimate [95% CI]          | <i>p</i> |
|--------------|------------|---------------------------|----------------------------|----------|
| primary      | same root  | model-dyadic <sup>†</sup> | +0.133 [+0.118, +0.148]    | <0.001   |
| primary      | same root  | root-dyadic               | +0.133 [+0.103, +0.162]    | <0.001   |
| primary      | same root  | delete-one-root jackknife | +0.133 [+0.108, +0.158]    | <0.001   |
| primary      | same month | model-dyadic <sup>†</sup> | +0.0039 [-0.0029, +0.0106] | 0.263    |
| primary      | same month | root-dyadic               | +0.0039 [-0.0043, +0.0120] | 0.353    |
| primary      | same month | delete-one-root jackknife | +0.0039 [-0.0048, +0.0125] | 0.383    |
| primary, S2  | same root  | model-dyadic <sup>†</sup> | +0.120 [+0.105, +0.135]    | <0.001   |
| primary, S2  | same root  | root-dyadic               | +0.120 [+0.100, +0.140]    | <0.001   |
| primary, S2  | same root  | delete-one-root jackknife | +0.120 [+0.098, +0.142]    | <0.001   |
| primary, S2  | same month | model-dyadic <sup>†</sup> | -0.0145 [-0.0268, -0.0022] | 0.021    |
| primary, S2  | same month | root-dyadic               | -0.0145 [-0.0279, -0.0011] | 0.034    |
| primary, S2  | same month | delete-one-root jackknife | -0.0145 [-0.0303, +0.0013] | 0.072    |
| expanded     | same root  | model-dyadic <sup>†</sup> | +0.127 [+0.115, +0.140]    | <0.001   |
| expanded     | same root  | root-dyadic               | +0.127 [+0.096, +0.159]    | <0.001   |
| expanded     | same root  | delete-one-root jackknife | +0.127 [+0.110, +0.145]    | <0.001   |
| expanded     | same month | model-dyadic <sup>†</sup> | +0.0059 [-0.0003, +0.0121] | 0.061    |
| expanded     | same month | root-dyadic               | +0.0059 [-0.0020, +0.0138] | 0.142    |
| expanded     | same month | delete-one-root jackknife | +0.0059 [-0.0023, +0.0142] | 0.160    |
| expanded, S2 | same root  | model-dyadic <sup>†</sup> | +0.122 [+0.111, +0.133]    | <0.001   |
| expanded, S2 | same root  | root-dyadic               | +0.122 [+0.096, +0.148]    | <0.001   |
| expanded, S2 | same root  | delete-one-root jackknife | +0.122 [+0.105, +0.139]    | <0.001   |
| expanded, S2 | same month | model-dyadic <sup>†</sup> | -0.0089 [-0.0185, +0.0008] | 0.071    |
| expanded, S2 | same month | root-dyadic               | -0.0089 [-0.0198, +0.0020] | 0.111    |
| expanded, S2 | same month | delete-one-root jackknife | -0.0089 [-0.0218, +0.0040] | 0.178    |

**TABLE VI: Outcome and weighting variants** (dyadic cluster-robust 95% CIs). Same month: 0-month gap bin relative to 12 or more months.

| Sample   | Variant                                                  | Shared root             | Same month                 | <i>p</i> (same month) |
|----------|----------------------------------------------------------|-------------------------|----------------------------|-----------------------|
| Primary  | primary outcome, OLS (pre-registered)                    | +0.133 [+0.118, +0.148] | +0.0039 [-0.0029, +0.0106] | 0.263                 |
| Primary  | WLS, weight = jointly-wrong items                        | +0.142 [+0.125, +0.159] | +0.0117 [+0.0016, +0.0217] | 0.023                 |
| Primary  | pairs with $\geq 1000$ jointly-wrong items               | +0.133 [+0.118, +0.148] | +0.0039 [-0.0029, +0.0106] | 0.263                 |
| Primary  | phi of error indicators (secondary)                      | +0.170 [+0.149, +0.191] | +0.0184 [+0.0097, +0.0272] | <0.001                |
| Primary  | agreement minus option-distribution chance (exploratory) | +0.125 [+0.111, +0.140] | -0.0018 [-0.0046, +0.0011] | 0.227                 |
| Expanded | primary outcome, OLS (pre-registered)                    | +0.127 [+0.115, +0.140] | +0.0059 [-0.0003, +0.0121] | 0.061                 |
| Expanded | WLS, weight = jointly-wrong items                        | +0.136 [+0.122, +0.151] | +0.0114 [+0.0033, +0.0195] | 0.006                 |
| Expanded | pairs with $\geq 1000$ jointly-wrong items               | +0.127 [+0.115, +0.140] | +0.0059 [-0.0003, +0.0121] | 0.061                 |
| Expanded | phi of error indicators (secondary)                      | +0.163 [+0.146, +0.180] | +0.0194 [+0.0118, +0.0270] | <0.001                |
| Expanded | agreement minus option-distribution chance (exploratory) | +0.116 [+0.105, +0.127] | +0.0007 [-0.0019, +0.0033] | 0.604                 |

**TABLE VII: Item-difficulty-aware nulls** (exploratory, added after review). Mean: average outcome across pairs. Estimates with 95% CIs and *p*-values from the delete-one-root jackknife.

| Sample       | Outcome                                   | Mean   | Shared root             | <i>p</i> | Same month                 | <i>p</i> |
|--------------|-------------------------------------------|--------|-------------------------|----------|----------------------------|----------|
| Primary      | Same-wrong agreement (primary outcome)    | 0.423  | +0.133 [+0.108, +0.158] | <0.001   | +0.0039 [-0.0048, +0.0125] | 0.383    |
| Primary      | N1a: minus item distractor null           | 0.002  | +0.133 [+0.107, +0.159] | <0.001   | +0.0040 [-0.0047, +0.0127] | 0.367    |
| Primary      | N1b: as N1a, roots weighted equally       | 0.019  | +0.133 [+0.108, +0.159] | <0.001   | +0.0038 [-0.0049, +0.0124] | 0.393    |
| Primary      | N2: co-failure over Rasch null (per item) | -0.000 | +0.040 [+0.032, +0.048] | <0.001   | +0.0036 [+0.0012, +0.0060] | 0.003    |
| Primary, s2  | Same-wrong agreement (primary outcome)    | 0.551  | +0.120 [+0.098, +0.142] | <0.001   | -0.0145 [-0.0303, +0.0013] | 0.072    |
| Primary, s2  | N1a: minus item distractor null           | -0.000 | +0.119 [+0.096, +0.142] | <0.001   | -0.0138 [-0.0310, +0.0034] | 0.116    |
| Primary, s2  | N1b: as N1a, roots weighted equally       | 0.023  | +0.120 [+0.097, +0.143] | <0.001   | -0.0143 [-0.0315, +0.0030] | 0.104    |
| Primary, s2  | N2: co-failure over Rasch null (per item) | -0.000 | +0.035 [+0.028, +0.042] | <0.001   | +0.0008 [-0.0019, +0.0036] | 0.561    |
| Expanded     | Same-wrong agreement (primary outcome)    | 0.434  | +0.127 [+0.110, +0.145] | <0.001   | +0.0059 [-0.0023, +0.0142] | 0.160    |
| Expanded     | N1a: minus item distractor null           | 0.004  | +0.128 [+0.110, +0.146] | <0.001   | +0.0061 [-0.0024, +0.0146] | 0.160    |
| Expanded     | N1b: as N1a, roots weighted equally       | 0.024  | +0.128 [+0.110, +0.146] | <0.001   | +0.0062 [-0.0023, +0.0146] | 0.154    |
| Expanded     | N2: co-failure over Rasch null (per item) | -0.000 | +0.038 [+0.032, +0.045] | <0.001   | +0.0036 [+0.0012, +0.0060] | 0.003    |
| Expanded, s2 | Same-wrong agreement (primary outcome)    | 0.547  | +0.122 [+0.105, +0.139] | <0.001   | -0.0089 [-0.0218, +0.0040] | 0.178    |
| Expanded, s2 | N1a: minus item distractor null           | -0.000 | +0.122 [+0.105, +0.140] | <0.001   | -0.0088 [-0.0226, +0.0051] | 0.214    |
| Expanded, s2 | N1b: as N1a, roots weighted equally       | 0.019  | +0.123 [+0.106, +0.141] | <0.001   | -0.0088 [-0.0229, +0.0052] | 0.216    |
| Expanded, s2 | N2: co-failure over Rasch null (per item) | -0.000 | +0.037 [+0.031, +0.043] | <0.001   | +0.0016 [-0.0028, +0.0060] | 0.480    |

## E. EXPLORATORY CONTROLS

Two checks were not pre-registered (TABLE VIII). Adding model size and same-architecture controls reduces the lineage estimate to 10.2–12.6 points; a shared architecture is

associated with a further 0.3–1.8 points. Shared lineage and shared architecture are nested in this population (fine-tunes keep their base architecture), so these coefficients partition one association rather than test two independent ones.

**TABLE VIII: Exploratory controls** (not pre-registered; all include the S1 covariate). Entries: estimate (standard error).

| Sample       | Models | Shared root (root-clustered SE) | Shared root (+ size, architecture) | Same month (+ size, architecture) | Same architecture |
|--------------|--------|---------------------------------|------------------------------------|-----------------------------------|-------------------|
| Primary      | 590    | +0.1328 (0.0144)                | +0.1256 (0.0082)                   | +0.0071 (0.0050)                  | +0.0027 (0.0026)  |
| Primary, S2  | 343    | +0.1166 (0.0117)                | +0.1018 (0.0087)                   | -0.0205 (0.0110)                  | +0.0182 (0.0042)  |
| Expanded     | 928    | +0.1263 (0.0134)                | +0.1201 (0.0062)                   | +0.0115 (0.0045)                  | +0.0050 (0.0022)  |
| Expanded, S2 | 589    | +0.1200 (0.0129)                | +0.1085 (0.0057)                   | -0.0088 (0.0081)                  | +0.0181 (0.0024)  |

## VII. DISCUSSION

**What the results say.** Within this population, which open-weight model a fine-tune descends from is strongly associated with which wrong answers it gives: pairs sharing a root agree on wrong answers about 13 points more often, a 31% increase over the mean. The time at which a model was uploaded adds little, and nothing of consistent sign, to the choice of wrong option. This is consistent with Kim *et al.*'s finding that a shared provider and architecture predict agreement [1], and refines it: with lineage taken from declared ancestry and time measured directly, the association runs through lineage.

**What they do not say.** The association is observational. Fine-tunes of one base share its parameters, its pretraining data and often similar post-training data; the design does not separate these routes, and it does not show that fine-tuning causes inherited errors. Nor does it show that release time is irrelevant: the pretraining era of a foundation model is a property of its root and therefore nested in lineage here, and the phi analysis suggests a small association between release proximity and which items fail together.

**Implications.** For practitioners combining open-weight models, the results suggest that drawing members from different base models is associated with less shared wrong-answer behavior than drawing them from one base at different times. We did not evaluate ensembles directly, and the effect on ensemble accuracy depends on how members are combined [2].

## VIII. LIMITATIONS

- 1) **Population.** The leaderboard population is self-selected and dominated by community fine-tunes of a few dozen bases between 2022 and 2024. Results may not hold for frontier or closed models, or for separately pretrained models released at the same time.
- 2) **Era definition.** Release time is the Hub creation month of the fine-tune, not the time its data were collected; pretraining era cannot be separated from lineage in this design.
- 3) **Lineage measurement.** Declared `base_model` fields can be wrong or missing; models with undeclared lineage are excluded from the primary population, and the expanded population relies on configuration paths that are weaker evidence. Conclusions agree across primary, expanded and strict populations.
- 4) **Benchmark.** One benchmark (MMLU, four options) is used. Many models are near chance; the S1 and S2 analyses and the item-level nulls address this, but preferences over answer options remain a concern for any agreement measure on multiple-choice tests.

- 5) **Evaluation pipeline.** All predictions were produced by the leaderboard's evaluation pipeline (the LM Evaluation Harness, 5-shot MMLU, log-likelihood scoring), not by us. Model-specific evaluation-configuration errors, including prompt or formatting mismatches, cannot be ruled out and may affect the observed agreement measures. Near-chance accuracy (339 of 977 models below 0.30) does not by itself show a configuration error—many are small or weakly trained models—but it cannot exclude one; the S2 analyses restrict to models above 0.30.
- 6) **Null and population dependence.** Following [23], levels of excess agreement depend on the null model and on the models and items compared; we report a contrast that is stable across the nulls examined, but other populations or benchmarks may give different contrasts.
- 7) **Contamination.** If MMLU items appear in training data, agreement may partly reflect memorization shared by lineage.
- 8) **Concentration.** The lineage contrast comes from 63 roots with two or more models in the primary sample (108 expanded); root-level inference accounts for this, but a few large roots carry substantial weight.

## IX. REPRODUCIBILITY

All code, frozen rosters (with SHA-256 hashes), Hub metadata caches, validation reports and analysis outputs are in the project repository at <https://github.com/SudharsanSelvaraj/Identifiable-Variance-Decomposition-of-Correlated-Errors-in-Foundation-Models>. A single script (`scripts/run_exp04_final.py`) re-runs the twelve pre-registered analyses, the inference and outcome audits and the exploratory checks from the frozen data, verifies the roster hashes, and diffs every coefficient against the committed outputs (maximum absolute difference below  $10^{-16}$ ); all tables in Sections IV–VI are generated from those outputs by `scripts/make_exp04_tables.py`. The leaderboard's per-item details declare no licence, so the derived item-level answer files (103 MB) are not redistributed; they are regenerated from the public source by `scripts/download_ollb_v1.py` from the public leaderboard and validated by `scripts/validation_report.py`. The pre-registration and its three dated amendments are in `docs/05_Experiments/Exp04_Leaderboard_Lineage_Era.md`.

## X. CONCLUSION

With the few families and quarters that real model populations offer, per-model variance decomposition cannot estimate the relative contributions of lineage and release era precisely enough for a reliable comparison, even when the variance components are formally identified. Moving to pairs of models, with lineage taken from declared ancestry and release time measured in months, makes the question estimable. Across 977 validated open-weight models, a shared lineage root is associated with a 12–13 point higher chance of choosing the same wrong answer when both models err, robustly across twelve pre-registered analyses and the inference, outcome and control variants examined. The association holds against item-difficulty-aware null models. Release-month proximity shows no association of consistent sign with the choice of wrong answer; it shows a small positive association with which items fail together in the full samples, which does not persist among models scoring at least 30%. Within this population, lineage, not release timing, is the dominant correlate of shared wrong-answer choices.

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## APPENDIX A CHANGES FROM THE EARLIER VERSION

The earlier version of this manuscript (archived in the repository as `paper/src/archive/`) reported a sixteen-model evaluation and an identifiability gate. The following findings of that version are withdrawn or corrected.

- 1) **Gate.** Its gate tested the rank, condition number and variance inflation of a fixed-effects dummy design. For the random-effects model it fitted, the variance components were identified (covariance-basis rank 3/3); the condition number depended on the reference level (115–585 for the same fifteen-model design). The claim that the variance components were “not identifiable” is withdrawn; the correct conclusion, imprecision, is in Section III.
- 2) **Item-level outputs.** The evaluation code recorded only the first option’s log-likelihood, so every per-question prediction was option A. The code is fixed and tested; those outputs are not used. The aggregate accuracy reported for Qwen-7B (0.2294) equalled the all-A rate and is withdrawn.
- 3) **Text errors.** Statements that a 30-model design “passes” at a condition-number threshold of 50 ( $\kappa = 93$ ), that the design sweep enforced crossing and connectedness by construction (100 of 105 configurations were crossed and 60 connected), that a linear-

probability model and a GLMM both reached the era boundary in 20–60% of replicates (only the GLMM did; the linear model 0–6.7%), and that a nested-design detector had “zero silent coverage” (the conditional quantity was undefined because no replicate went undetected) are corrected.

- 4) **Population history.** Four models in the sixteen-model study’s pre-registered plan (Llama-3.1, Llama-3.3, Qwen1.5 and Phi-4-reasoning-vision-15B) were not evaluated, without a recorded reason; the two DeepSeek models were planned for imputation. That arm is no longer used for inference.

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